Joint Mean-Variance Inference for Nonlinear Mixed-Effects Models with Measurement Errors and Outliers

Qian Ye

University of British Columbia

qian.ye@stat.ubc.ca

July 27, 2022

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Overview

Motivation
- Motivating data set
- NLME limitaions
- Jointly model mean and variance

The Models
- The mean and variance models for NLME
- Robust model
- Measurement error model

Joint Likelihood Inference
- Joint likelihood
- Estimation

DTP Data Analysis
- The data set
- Methods
- Results

Simulations
- Design
- Results

Motivating data set - DTP data

The Drug Treatment Program (DTP) in the British Columbia Centre for Excellence (BCCfE) provides the longitudinal viral load measurements during antiretroviral treatments (ART).

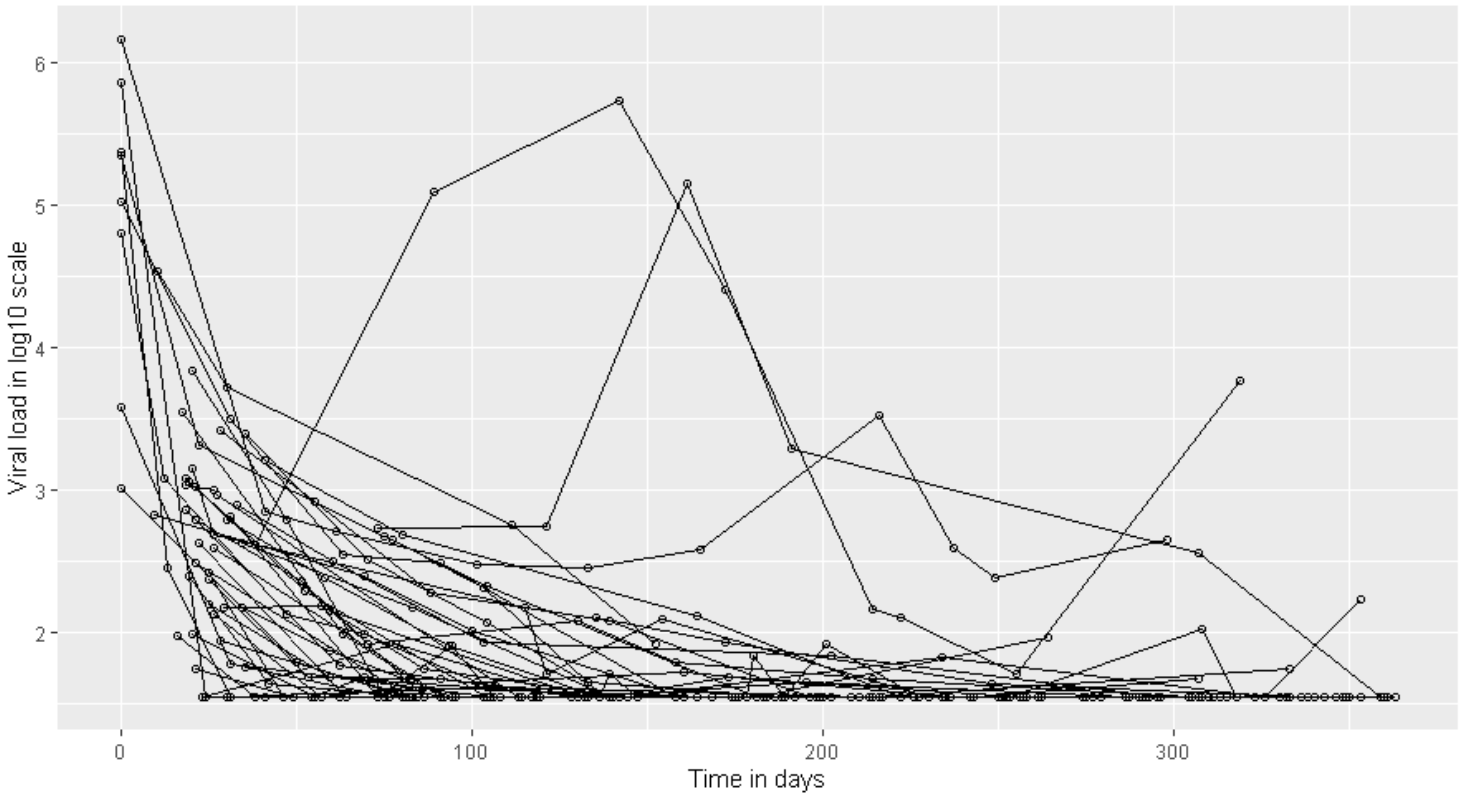
Some research questions motivated by the data set

1. Studying the longitudinal virologic dynamics since ART initiation
2. Evaluating and comparing efficacy between different regimens

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Viral load trajectories

Figure: Viral Load (log10-) trajectories within the first year



Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Feature of the data

- Viral loads decline rapidly in the initial period, roughly following an exponential pattern.
  (may use non-linear mixed-effects (NLME) model)
- The variations of the within-individual repeated measurements seems not constant over time and across individuals
- Potential outliers: some data do not seem to follow the general patterns of the whole data

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

NLME

NLME models may be more desirable than empirical models (e.g., linear mixed-effects (LME) models):

- Describe the data-generation mechanisms
- Parameters often provide meaningful and justified interpretations (e.g., initial decay rate in exponential decay NLME[1].
- Fewer parameters and simpler form to provide a better fit of data
- Provide more reliable predictions for missing or future values.

But, ...

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

The limitations in the standard NLME

The standard NLME (i.e., the method implemented in `nlme` R package[2])

- assume constant within-individual variance for all individuals
- may be greatly affected by outliers

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Jointly model mean and variance for NLME

The innovation in our research:

In addition to modeling the mean based on an NLME model, we also model the within-individual variance with

- time-varying covariates
- individual-specific random effects

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Advantages

Importance of modeling the variance:

- Obtain more efficient estimates (i.e., with smaller SE) of the main parameters of interest
- Avoid bias in estimation
- Robust inference to address outliers in the data

E.g., model the variance in the exponential decay model

$$y_{ij} = \beta_{1i} + \beta_{2i} e^{-\beta_{3i} t_{ij}} + e_{ij}$$

Obtain an more efficient and accurate estimate of initial viral decay rates (i.e. more accurate estimate of β_3 with smaller SE).
Better evaluate and compare different treatments since the initial viral decay rates may reflect the efficacies of the treatments[1]. (DTP data analysis in Section 4)

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Other issues in the data

Measurement errors in time-varying covariates:

In the variance model, the covariate may be measured with errors. The covariate effect on the variance could be masked if the measurement errors are not addressed.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Nonlinear mixed-effects (NLME) models

NLME model - the mean model

$$y_{ij} = g(\boldsymbol{x}_{ij}, \boldsymbol{\beta}, \boldsymbol{u}_i) + e_{ij} \tag{1}$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n_i$, where

- $g(\cdot)$: a known nonlinear function for the mean $\mathrm{E}(y_{ij})$
- $\boldsymbol{\beta}$: fixed effects; $\boldsymbol{u}_i$: random effects.
- e_{ij}: within-individual error. $\cancel{e_{ij} \sim \mathcal{N}(0, \sigma_0^2)}$

$$e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$$

allow the within-individual variances to be <u>individual-specific</u> and <u>time-dependent</u>

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Nonlinear mixed-effects (NLME) models

Model the within-individual variances σ_{ij}^2 with time-varying covariates and random effects.

NLME model - the variance model

$$\log(\sigma_{ij}^2) = \alpha_1 + \alpha_2 z_{ij} + a_i \tag{2}$$

for $i = 1, \cdots, n$; $j = 1, \cdots, n_i$, where

- z_{ij}: a time-varying covariates
- a_i: a individual-specific random effect. Assume $a_i \sim \mathcal{N}(0, \sigma_a^2)$ or $\exp(a_i) \sim k/\chi_k^2$

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Re-cap the advantages of modeling the variance

- Better understand the nature of the within-individual variation
- Improve efficiency and accuracy in estimation
- Robust inference to address outliers in the data

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Robust model

In the variance model (2), if we assume $\exp(a_i) \sim k/\chi^2_k$, it can be derived that

the within-individual error $e_{ij} \sim t(k)$,

where $t(k)$ is a t-distribution with degrees of freedom k.[3]

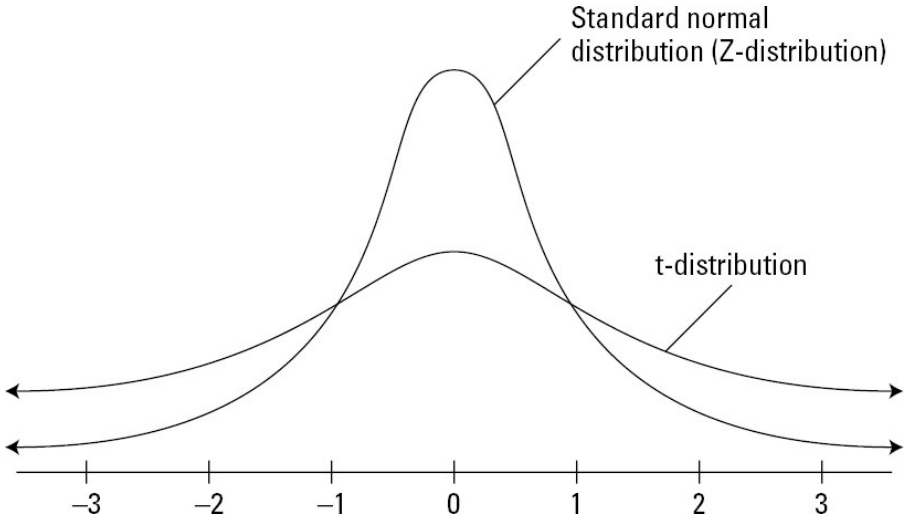


Figure: $t(k)$ has heavier tails, and thus it incorporates possible outliers

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Measurement error model

To address the measurement error in covariate z_{ij}, model the covariate process using an empirical mixed-effects model.

Measurement error model

$$z_{ij} \equiv z_{ij}^* + \epsilon_{ij} = q(t_{ij}, \gamma, \boldsymbol{b}_i) + \epsilon_{ij} \tag{3}$$

- z_{ij}: observed covariate value; z_{ij}^*: unobserved true covariate value; ϵ_{ij}: measurement error
- $q(\cdot)$: a known linear function or a unknown smooth function[4]
- γ: fixed effects
- $\boldsymbol{b}_i$: random effects

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Joint likelihood

For inference, we can use the joint likelihood of all models and obtain MLEs of all parameters by maximizing the joint likelihood,

$$L = \prod_{i=1}^{n} \int \int \int f(\mathbf{y}_i | \mathbf{x}_i, \mathbf{z}_i, \boldsymbol{\beta}, D, \boldsymbol{\sigma}_i^2, \mathbf{u}_i) f(\boldsymbol{\sigma}_i^2 | \boldsymbol{\alpha}, a_i) \\ f(\mathbf{z}_i | \boldsymbol{\gamma}, \mathbf{b}_i) f(\mathbf{u}_i, \mathbf{b}_i, a_i | \boldsymbol{\theta}) \, d\mathbf{u}_i \, da_i \, d\mathbf{b}_i.$$

The computation is extremely intensively:

- high-dimensional and intractable integrals due to unobservable random effects
- convergence problems especially when nonlinear models are used and missing data or measurement errors are present [5].

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

H-likelihood approximation

A computationally more efficient approximate method for approximate likelihood inference, using the h-likelihood[6],

$$h = \sum_{i=1}^{n} \{\log f(\mathbf{y}_i | \, \mathbf{x}_i, \mathbf{z}_i, \boldsymbol{\beta}, D, \boldsymbol{\sigma}_i^2, \mathbf{u}_i) +$$

$$\log f(\boldsymbol{\sigma}_i^2 | \boldsymbol{\alpha}, a_i) + \log f(\mathbf{z}_i | \boldsymbol{\gamma}, \mathbf{b}_i) + \log f(\mathbf{u}_i, \mathbf{b}_i, a_i | \boldsymbol{\theta}).\}$$

There is no high-dimensional and intractable integrals in the h-likelihood as the random effects $\mathbf{v}_i = (\mathbf{u}_i, \mathbf{b}_i, a_i)$ are treated as "parameters".

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Estimation procedure

Notations:

- $\mathbf{v}_i = (\mathbf{u}_i, \mathbf{b}_i, a_i)$: random effects in all models
- $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\gamma})$: all fixed mean parameters
- $\boldsymbol{\xi}$: all fixed variance-covariance parameters

Starting with some initial values $(\mathbf{v}^{(0)}, \boldsymbol{\theta}^{(0)}, \boldsymbol{\xi}^{(0)})$, at iterative $k, k = 0, 1, 2, \cdots$,

- Step 1: Solve $\frac{\partial h}{\partial \mathbf{v}} = 0$ and obtain updated estimate of the random effects $\mathbf{v}^{(\mathbf{k+1})}$;
- Step 2: Solve $\frac{\partial h}{\partial \boldsymbol{\theta}} = 0$ and obtain updated estimates of the mean parameters $\boldsymbol{\theta}^{(k+1)}$;
- Step 3: Solve $\frac{\partial h}{\partial \boldsymbol{\xi}} = 0$ and obtain updated estimates of the variance-covariance parameters $\boldsymbol{\xi}^{(k+1)}$.

Iterate the above steps until convergence.[7]

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Real data analysis - DTP data

We are interested in

1. Modeling the longitudinal virologic dynamics since ART initiation using exponetial decay NLME
2. Evaluating and comparing efficacy between different regimens by examining the initial viral decay rates

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Inclusion/Exclusion

The analytic data set includes

1. patients initiated treatment after April 01, 2011
2. follow-up time within the first year since the ART initiation before rebound
3. patients with 4 more more measurements in both VL and CD4

Result in 1796 patients with 11904 records in total.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Sample description

- Median age of 40 at ART initiation; 87% males; 18% IDU
- 99% viral suppressed; 13% rebounded after suppression
- Median follow-up of 320 days (range from 73-365 days); re-scale time to $[0, 1]$
- The pharmacological classes within the first regimen

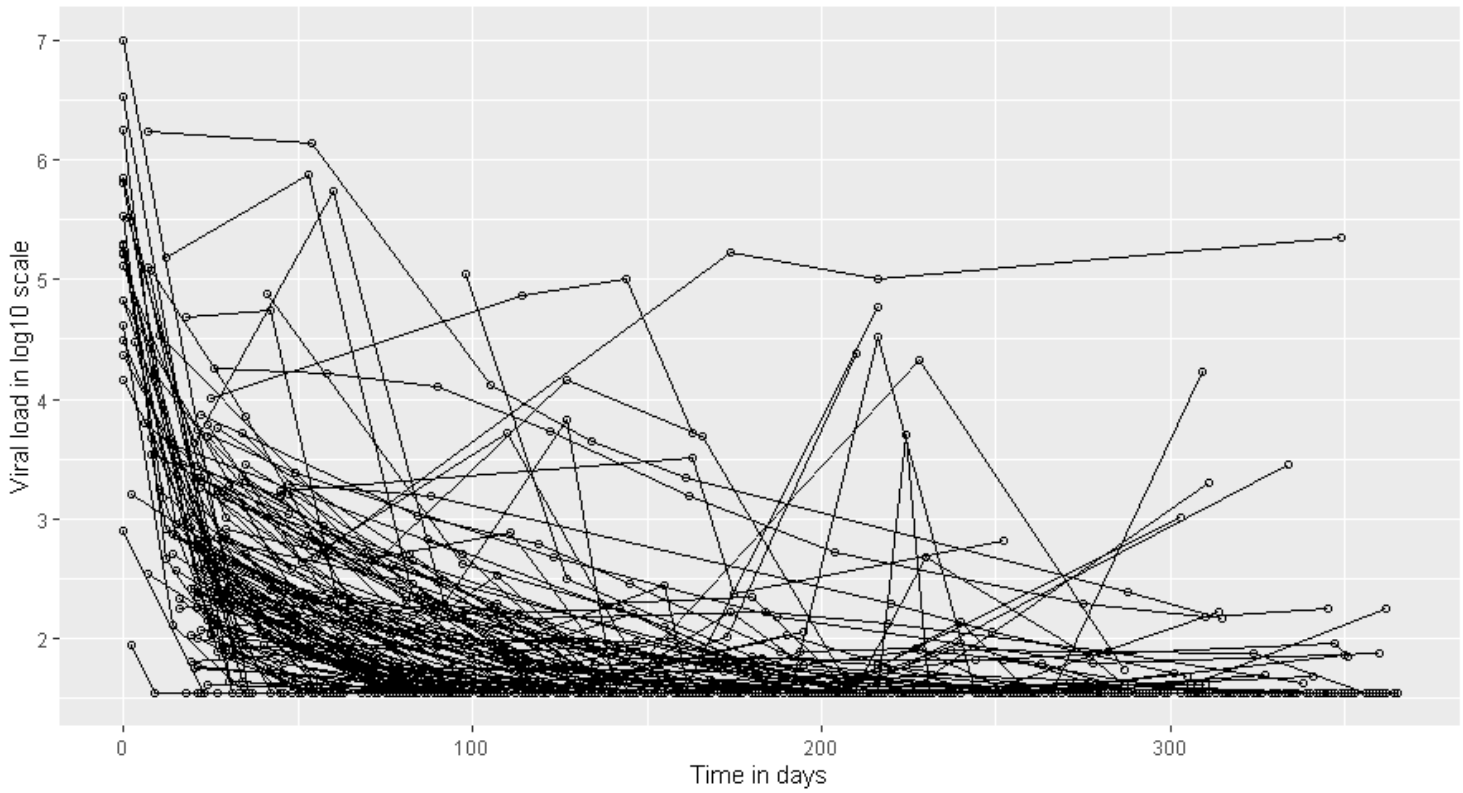
First regimen class	N	%
CAN/	5	0.28
IIN/	302	15.82
IIN/ENH/	113	6.29
NNR/	506	28.17
PI/ENH/	729	40.59
PI/IIN/ENH//	141	7.85

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Viral load trajectories

Median number of VL measurements is 6 per patient, ranging from 4-15

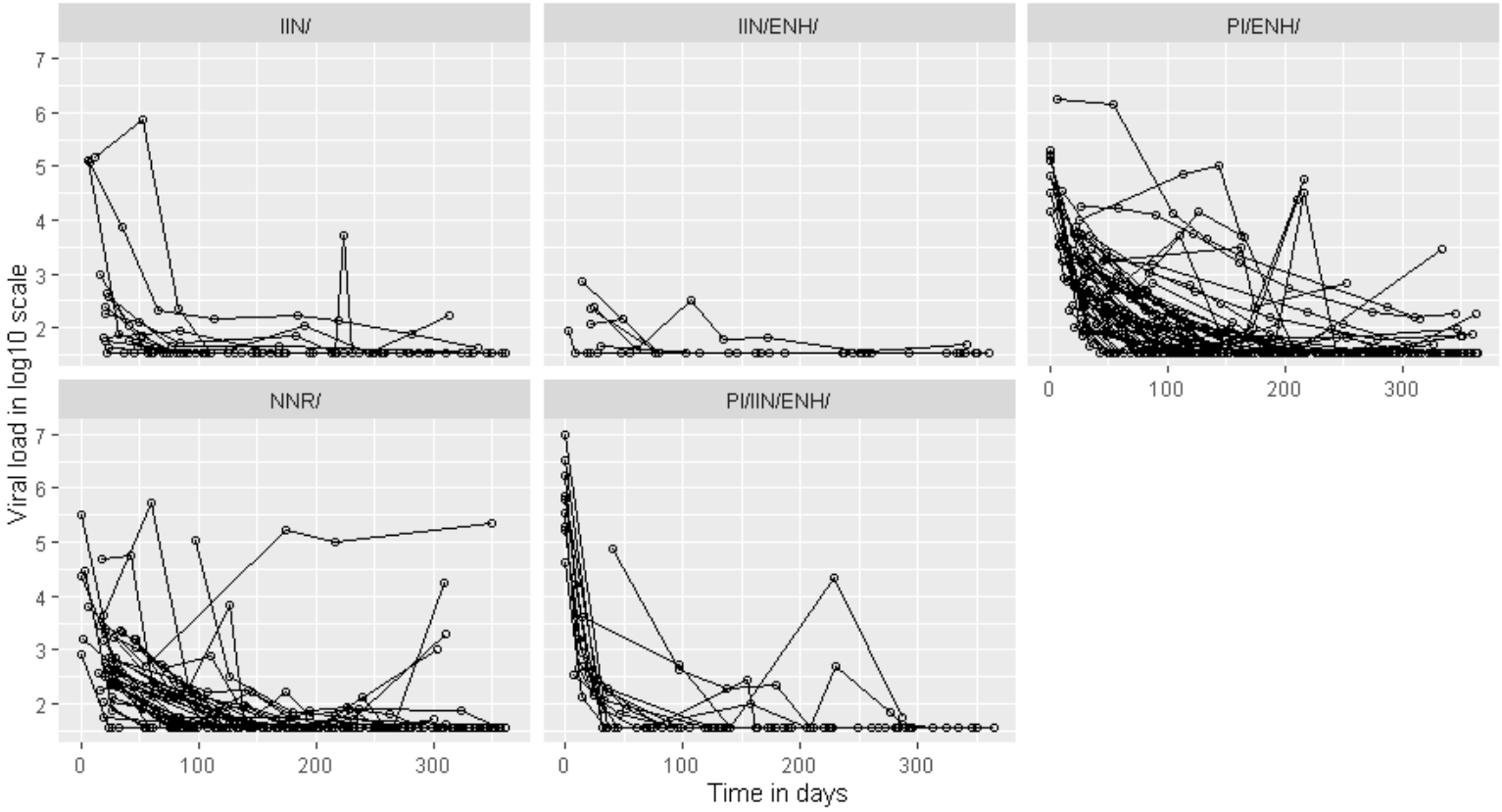
Figure: Viral load (log10-) trajectories for 150 randomly selected patients



Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Viral load trajectories by regimen class

Figure: Viral load (log10-) trajectories for 150 randomly selected patients, by first regimen class



Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Analytic plan

Use the proposed models to explore the initial viral decay rate

- for each regimen class
- modeling the mean using one-compartment exponential decay NLME model[1]
- modeling the within-individual variances with time-varying CD4 and random effect a_i
- assuming $\exp(a_i) \sim k/\chi^2_k$ in the variance model to address possible outliers in the viral load measurements
- considering measurement error model to address the measurement errors in the CD4 measurements

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Models

NLME - the mean model

$$y_{ij} = (\beta_1 + u_{i1}) + (\beta_2 + u_{i2})e^{-(\beta_3 + u_{i3})t_{ij}} + e_{ij}$$

- y_{ij}: the $\log_{10}$ viral load measured at time t_{ij} for individual i
- β_3: initial viral decay rate
- $e_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$

NLME - the variance model

$$\log(\sigma_{ij}^2) = \alpha_1 + \alpha_2 z_{ij}^* + a_i$$

- z_{ij}^*: the "true" but unobservable (log-) CD4
- $\exp(a_i) \sim k/\chi_k^2$ for robust inference (pre-specify $k = 3$)

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Models

Measurement error model for CD4

$$z_{ij} \equiv z_{ij}^* + \epsilon_{ij} = (\gamma_0 + b_{i0}) + (\gamma_1 + b_{i1})t_{ij} + (\gamma_2 + b_{i2})t_{ij}^2 + \epsilon_{ij}$$

- z_{ij}: the observed (log-) CD4 with measurement error
- $(\gamma_0, \gamma_1, \gamma_2)$: fixed effects
- (b_{i0}, b_{i1}, b_{i2}): random effects

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results: parameter estimates for two regimen classes

IIN/ENH regimen class:

Table: Model results for IIN/ENH regimen class

Model		IIN/ENH						
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1
Standard	Est	1.61	2.70	23.38				
	SE	0.027	0.154	2.166				
	p-val	$< .001$	$< .001$	$< .001$				
TS	Est	1.54	2.71	30.05	11.31	-2.41	6.08	0.22
	SE	0.005	0.056	1.451	0.699	0.113	0.062	0.043
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$
JM	Est	1.54	2.67	31.33	14.01	-2.89	6.06	0.28
	SE	0.003	0.047	1.590	0.744	0.121	0.061	0.032
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$

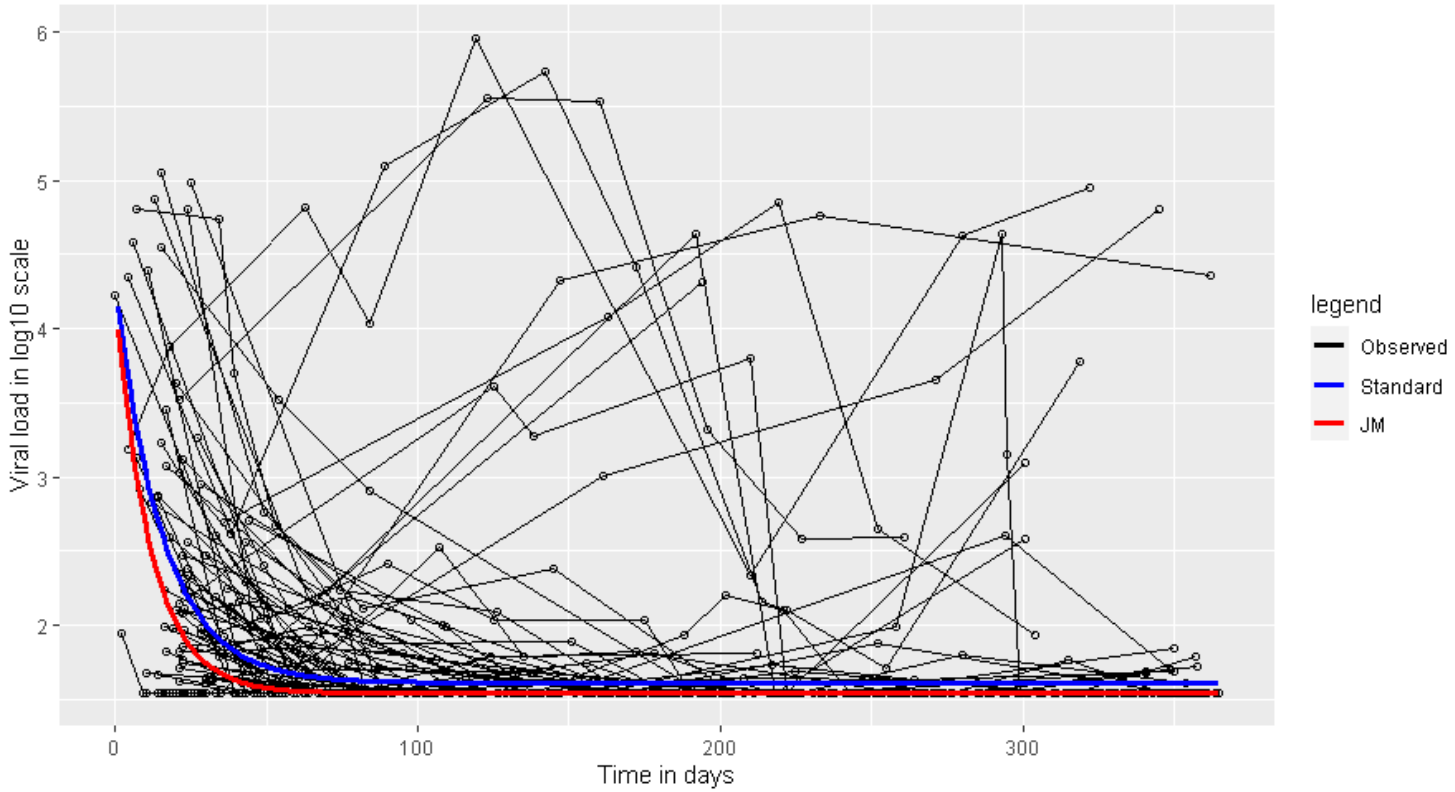
Standard: NLME without modeling variances. TS: two-step method: predict true CD4 first, and then fit NLME. JM: Joint modeling all models

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results: parameter estimates for two regimen classes

IIN/ENH regimen class:

Figure: IIN/ENH regimen class: fitted population average trajectory

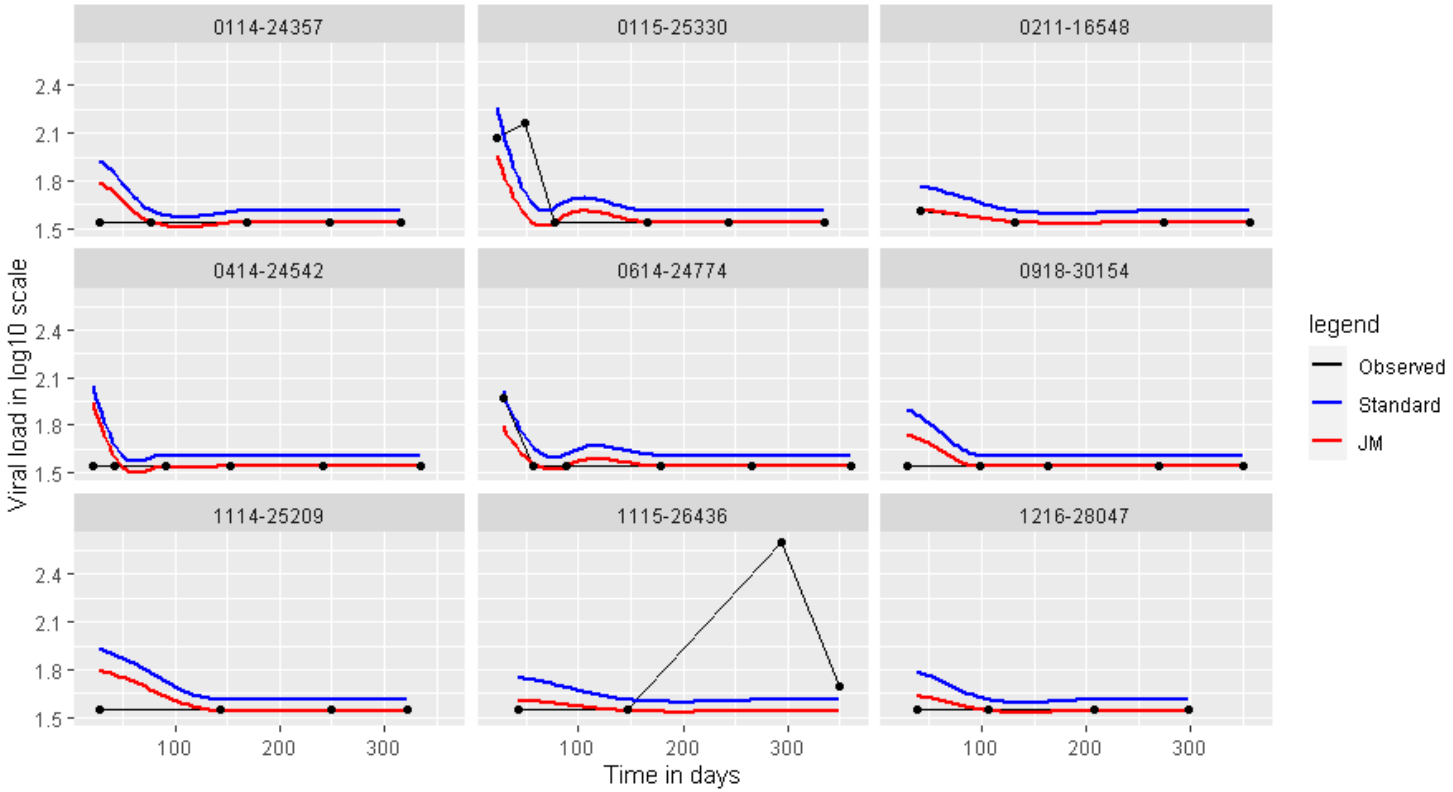


Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results: parameter estimates for two regimen classes

IIN/ENH regimen class:

Figure: IIN/ENH regimen class: Fitted trajectory for randomly selected patients



Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results: parameter estimates for two regimen classes

NNR/ regimen class:

Table: Model results for NNR/ regimen class

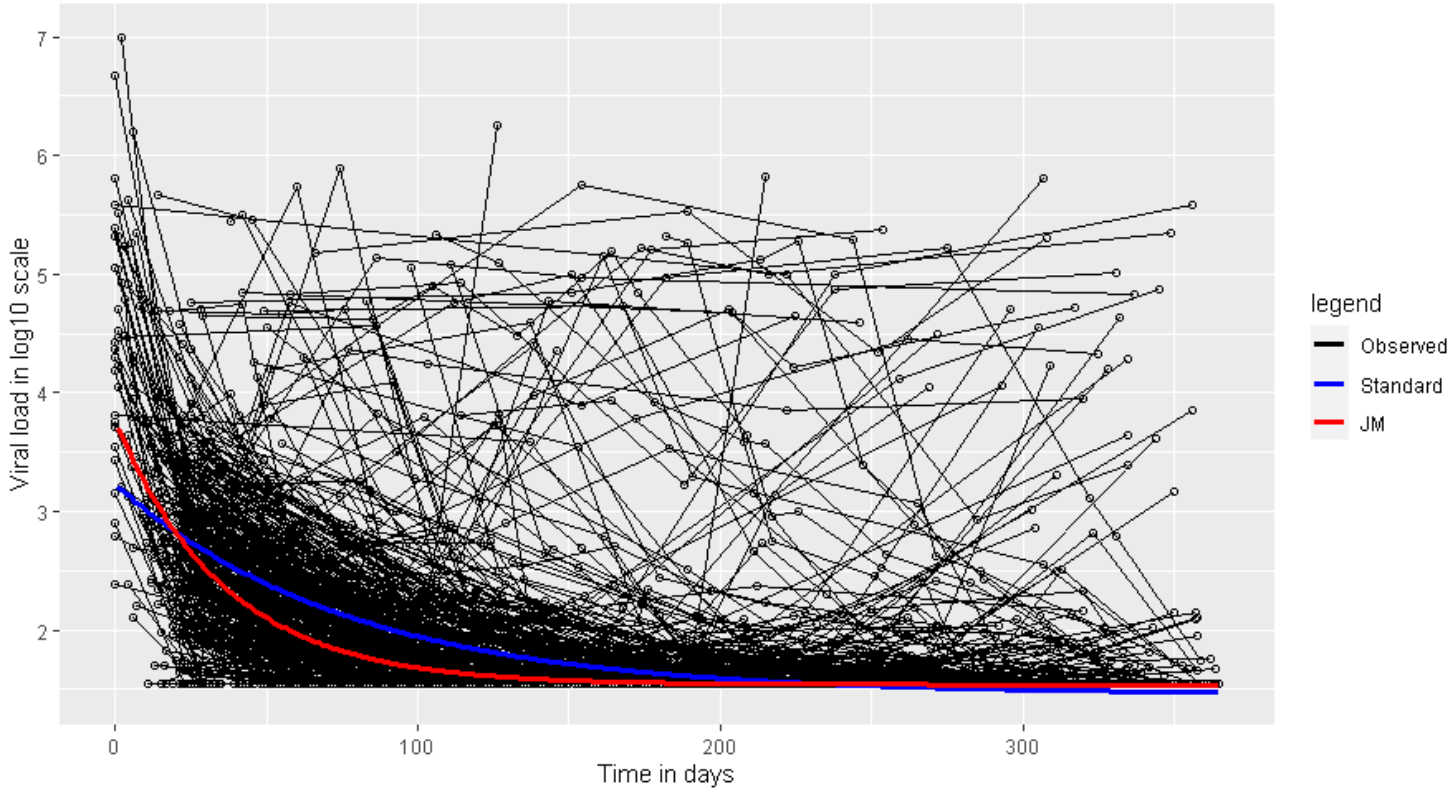
Model		NNR/							
		β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
Standard	Est	1.46	1.77	4.71					
	SE	0.013	0.063	0.263					
	p-val	$< .001$	$< .001$	$< .001$					
TS	Est	1.54	2.21	9.99	3.82	-1.16	5.89	0.69	-0.37
	SE	0.003	0.050	0.147	0.254	0.041	0.035	0.092	0.087
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	
JM	Est	1.54	2.22	10.03	4.44	-1.27	5.87	0.78	-0.43
	SE	0.003	0.050	0.149	0.250	0.041	0.032	0.056	0.053
	p-val	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	$< .001$	

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results: parameter estimates for two regimen classes

NNR/ regimen class:

Figure: NNR/ regimen class: fitted population average trajectory

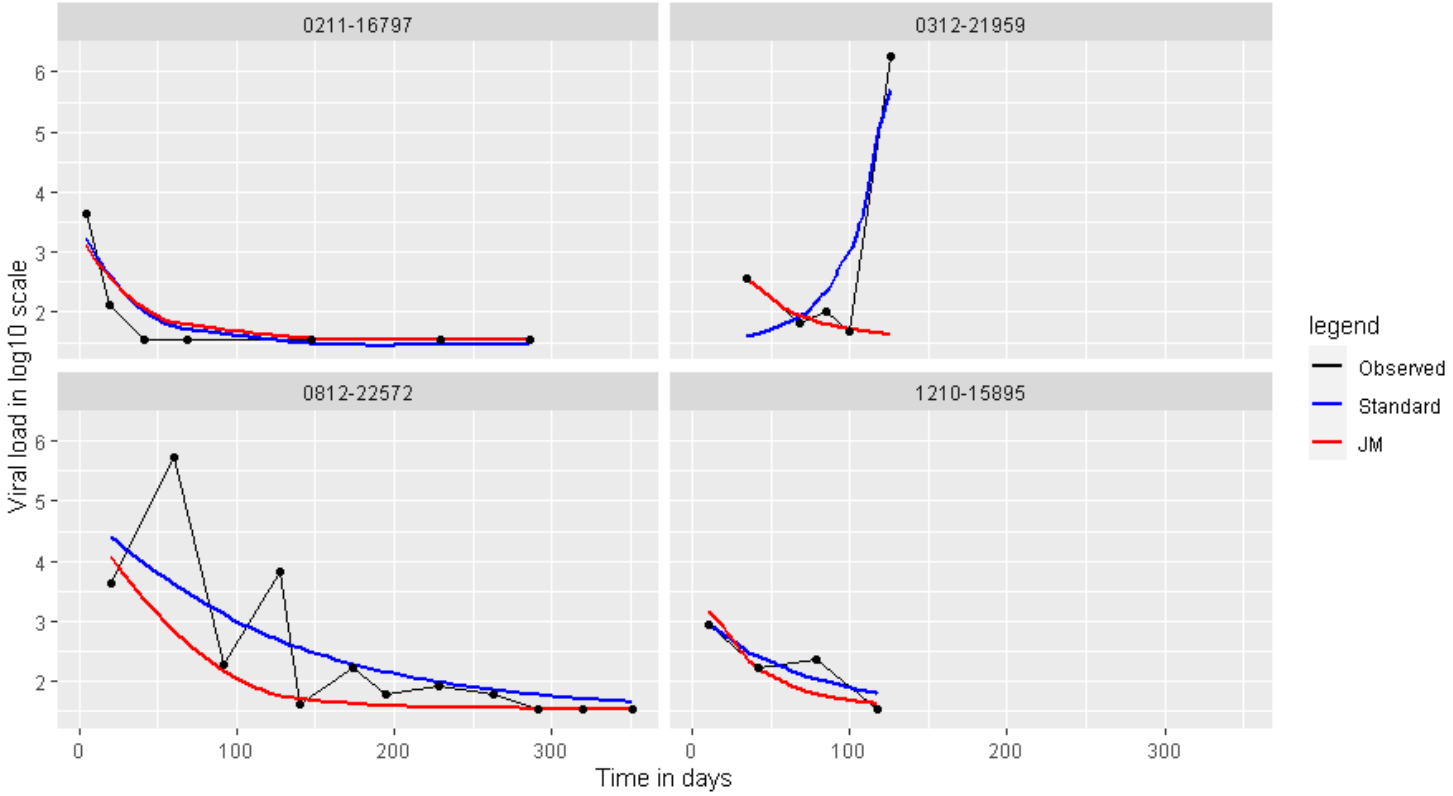


Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results: parameter estimates for two regimen classes

NNR/ regimen class:

Figure: NNR/ regimen class: Fitted trajectory for randomly selected patients



Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Simulations

To evaluate and compare the performance between Standard, TS, and JM, simulation study was conducted:

- simulated data set based on models with known parameters
- fitted the data set using Standard, TS, and JM separately
- compared the parameter estimates to the true values

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Simulation design

- n=100 individuals, with $n_i = 15$ repeated measurements for each individual.
- The entire study period is set to be [0, 1], and the 15 measuring time points are equally allocated
- The models and true parameter values are chosen based on real data analysis
- Consider two scenarios:
  - no outliers: generate within-individual errors using normal distribution
  - possible outliers: generate within-individual errors using $t(3)$ distribution
- The number of simulation repetition: 100

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Simulation design

Compare the parameter estimation between Standard, TS, and JM , in term of

- "Estimate": average of the 100 estimates from the 100 simulation runs
- "rMSE": relative mean square error of the 100 parameter estimates.

  rMSE (%) of $\hat{\beta} = \left| \frac{1}{100} \sum_{m=1}^{100} \left(\hat{\beta}^{(m)} - \beta \right)^2 / \beta \right| \times 100\%$
- "SE": square root of the average of the 100 variance estimates
- "Coverage":empirical coverage based on SE among the 100 95% confidence intervals from the 100 simulation runs.
- "SE.bt": the average of bootstrapping SE from the 100 simulation runs (For JM estimation only)
- "Coverage.bt": empirical coverage based on bootstrapping SE (For JM estimation only)

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results

Scenario 1: No outlier

Table: Normal distribution for a_i: both data generation and modeling

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.527	2.996	8.054					
	SE	0.067	0.148	0.862					
	rMSE(%)	0.235	0.441	14.313					
	Coverage	0.920	0.990	0.910					
TS	Estimate	2.509	2.994	7.699	-8.392	1.547	5.206	1.598	-1.198
	SE	0.052	0.069	0.433	0.492	0.087	0.042	0.063	0.061
	rMSE(%)	0.102	0.222	3.647	8.426	1.252	0.033	0.207	0.258
	Coverage	0.960	0.920	0.930	0.760	0.800	0.980	0.970	0.960
JM	Estimate	2.511	2.995	7.759	-8.069	1.489	5.205	1.603	-1.202
	SE	0.052	0.069	0.441	0.486	0.086	0.042	0.060	0.058
	SE. bt	0.057	0.075	0.495	0.690	0.122	0.043	0.061	0.059
	rMSE(%)	0.106	0.222	4.027	7.479	1.267	0.033	0.193	0.237
	Coverage	0.950	0.930	0.920	0.800	0.810	0.980	0.970	0.960
	Coverage.bt	0.970	0.960	0.950	0.930	0.920	0.970	0.960	0.970

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Results

Scenario 2: Possible outliers

Table: Inverse χ^2 distribution with 3 degree of freedom for $\exp(a_i)$: both data generation and modeling

Model	Parameter	β_1	β_2	β_3	α_1	α_2	γ_0	γ_1	γ_2
	True value	2.5	3.0	7.5	-8	1.5	5.2	1.6	-1.2
Standard	Estimate	2.490	3.007	7.546					
	SE	0.090	0.222	1.245					
	rMSE(%)	0.431	0.963	21.293					
	Coverage	0.920	0.990	0.950					
TS	Estimate	2.505	2.991	7.582	-8.024	1.552	5.199	1.605	-1.204
	SE	0.056	0.081	0.500	0.491	0.087	0.042	0.063	0.061
	rMSE(%)	0.162	0.242	5.636	6.973	1.368	0.043	0.254	0.328
	Coverage	0.920	0.940	0.850	0.810	0.820	0.940	0.960	0.960
JM	Estimate	2.507	2.991	7.645	-7.738	1.500	5.197	1.611	-1.208
	SE	0.055	0.081	0.507	0.486	0.086	0.042	0.060	0.058
	SE.bt	0.071	0.106	0.679	0.755	0.134	0.042	0.061	0.058
	rMSE(%)	0.163	0.241	6.023	8.892	1.363	0.043	0.245	0.316
	Coverage	0.910	0.940	0.850	0.740	0.780	0.940	0.920	0.910
	Coverage.bt	0.960	0.990	0.970	0.910	0.920	0.940	0.920	0.930

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Conclusion

In the DTP data analysis, we see that

- *Standard* and *JM* provided very different estimates of initial decay rate:
  23.38 vs. 31.33 for IIN/ENH/
  4.71 vs. 10.03 for NNR/
- By modeling the variance, *JM* (and *TS*) obtain more efficient estimates for the parameters of main interest. E.g., $\sim 50\%$ reduction in SE of β_3 compared to *Standard*.
- *JM* provide better fit to the data and less sensitive to outliers

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Conclusion

Based on the simulation results, we may conclude that

- *JM* and *TS* obtain less biased parameter estimates than *Standard*, especially in the case where the outliers present.
- *JM* and *TS* obtain more efficient estimates for the parameters of main interest, with much smaller SE
- *JM* and *TS* may underestimate the SE for some parameters, but the bootstrapping method provides more reliable SE for *JM*.

All in all, *JM* with bootstrapping-adjusted SE outperforms other methods, providing more reliable and efficient statistical inference.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Limitations

- Computation is still intensive, especially with bootstrapping
- Didn't address the data truncation by the lower limit of detection in viral load measures.
- The h-likelihood approximation to the joint likelihood may be less satisfactory with discrete responses such as binary responses [8]

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

Future Work

Joint with a survival outcome (e.g. time to dropout, time to rebound, etc), with linking the event time
i) to the random effects from the NLME model being used as "covariates", or
ii) to the unobserved true time dependent covariates,
to see if the individual-specific characteristics of the viral load trajectories may be predictive for the risk of event.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

The End

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

References I

[1] Hulin Wu and A. Adam Ding. "Population HIV-1 dynamics in vivo: Application models and inferential tools for virological data from AIDS clinical trials". In: Biometrics 55.2 (1999), pp. 410–418.

[2] M.J. Lindstrom and D.M. Bates. "Nonlinear mixed effects models for repeated measures data". In: Biometrics 46 (1990), pp. 673–687.

[3] Kenneth L. Lange, Roderick J. A. Little, and Jeremy M. G. Taylor. "Robust Statistical Modeling Using the t Distribution". English. In: Journal of the American Statistical Association 84.408 (1989), pp. 881–896.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

References II

[4] L. Wu, W. Liu, and J. Hu. "Joint inference on HIV viral dynamics and immune suppression in presence of measurement errors". In: Biometrics 66 (2010), pp. 327–335.

[5] L. Wu, X. J. Hu, and H. Wu. "Joint inference for nonlinear mixed-effects models and time-to-event at the presence of missing data.". In: Biostatistics 9 (2008), pp. 308–320.

[6] Y. Lee, J.A. Nelder, and Y. Pawitan. Generalized Linear Models with Random Effects: Unified Analysis via H-
2nd. London: Chapman & Hall/CRC, 2017.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References

References III

[7] E.F. Vonesh et al. "Conditional second-order generalized estimating equations for generalized linear and nonlinear mixed-effects models.". In: Journal of the American Statistical Association 97 (2002), pp. 271–283.

[8] H. Joe. "Accuracy of Laplace approximation for discrete response mixed models.". In: Computational Statistics & Data Analysis 52 (2008), pp. 5066–5074.

Motivation
Motivating data set
NLME limitaions
Jointly model mean and variance
The Models
The mean and variance models for NLME
Robust model
Measurement error model
Joint Likelihood Inference
Joint likelihood
Estimation
DTP Data Analysis
The data set
Methods
Results
Simulations
Design
Results
Conclusion and Discussion
References